

Worked for 38s

Yes. **A3 is where we make the HQ difficult to design incorrectly.** We now decide which departments must touch, which should remain close, which can be separated, and which should deliberately **not** have a direct route.

This step will heavily constrain A4, the actual floor plan.

A3 — Department Adjacency

A3.1 Adjacency language

I recommend four relationship grades:

Code	Relationship	Meaning for A4
A — Direct	Must directly connect	Shared doorway, controlled doorway, shared vestibule or immediate opening
B — Near	Should be physically close	Normally one short corridor/room away
C — Independent	No strong adjacency requirement	Can be placed where the wider layout works
D — Separate	Should not directly connect	Separation supports security, workflow or hierarchy

A **D** does not mean the departments cannot communicate digitally. It means we should not casually give them a direct physical door.

A3.2 The core adjacency matrix

Abbreviations:

ENT

Entrance / Reception / Identity

OPS

Central Operations

FND

Training & Foundations

COM

Communications

CRY

Cryptography Lab

SOC

Security Operations Centre

RT

Red Team Operations

ENG

Engineering

QNT

Quantum Wing

CMP

Advanced Compute

CORE

Secure Core

	ENT	OPS	FND	COM	CRY	SOC	RT
ENT	—	A	B	D	D	D	D
OPS	A	—	A	A	B	A	D
FND	B	A	—	C	B	C	D
COM	D	A	C	—	A	A	D
CRY	D	B	B	A	—	B	D
SOC	D	A	C	A	B	—	A
RT	D	D	D	D	D	A	—
ENG	D	B	B	B	B	C	C
QNT	D	D	D	B*	B	D	D
CMP	D	D	D	B	B	B	B

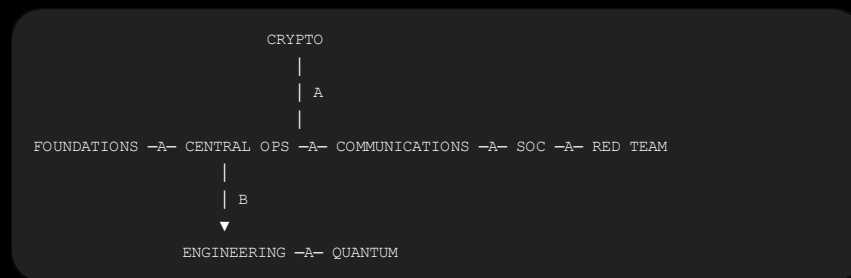
	ENT	OPS	FND	COM	CRY	SOC	RT
CORE	D	D	D	D	D	D	D

* means the relationship is technically important, but I would handle it through a **controlled technical connection/vestibule**, not an ordinary public doorway.

That distinction becomes important below.

A3.3 The primary structural chain

The strongest relationships create this backbone:



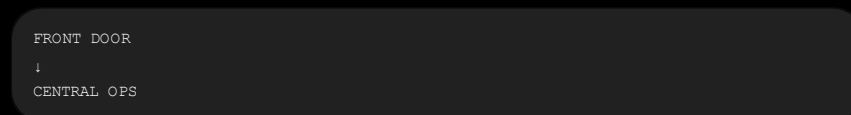
That is already much closer to a real floor-plan constraint.

However, it does **not** mean we draw those rooms in one straight line.

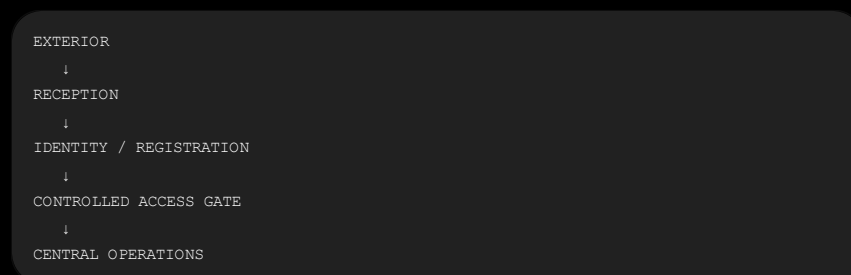
A3.4 Entrance ↔ Central Operations

This must be **A — Direct**.

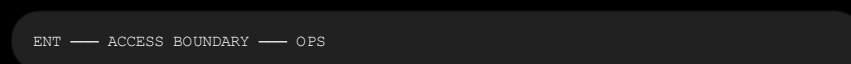
But not:



The physical sequence must remain:



So technically:



This is the most important arrival connection in the building.

It supports the Prologue perfectly and creates the first security threshold.

A3.5 Foundations ↔ Central Operations

Also **A — Direct**.

I think this should survive into the final plan.

The learner's first newly available department should be extremely easy to reach from their normal workstation environment.

Conceptually:

```
FOUNDATIONS
|
controlled door
|
CENTRAL OPERATIONS
```

During the Prologue:

```
FOUNDATIONS
ACCESS RESTRICTED
```

After PQ-001:

```
FOUNDATIONS
ACCESS GRANTED
```

The player physically sees their first progression unlock.

A3.6 Communications ↔ Central Operations

A — Direct.

Communications is too important to normal Phantom Q operations to bury in a specialist wing.

Central Ops needs quick access to:

```
communications state
sessions
transfer systems
remote participants
Bob
network activity
```

Therefore:

```
CENTRAL OPS
|
▼
COMMUNICATIONS
```

should be one of the clearest connections in HQ.

A3.7 Communications ↔ Cryptography

Also **A — Direct.**

This is one of the most meaningful adjacencies in the facility.

```
CRYPTOGRAPHY
|
protect information
|
▼
COMMUNICATIONS
|
move information
```

That physical relationship mirrors the conceptual relationship we teach.

Incidents 02–05 also repeatedly move between them.

So this should not be a coincidence in the final floor plan.

A3.8 Communications ↔ SOC

Again A — Direct.

This gives us an extremely strong functional chain:

```
CRYPTO
↓
COMMUNICATIONS
↓
SOC
```

Which represents:

```
PROTECT
↓
TRANSMIT
↓
MONITOR / INVESTIGATE
```

That is excellent architectural storytelling.

A3.9 Central Operations ↔ SOC

I would also keep this A — Direct, but through a **secure controlled boundary**.

So SOC effectively has two important interfaces:

```
CENTRAL OPS
|
| secure
|
▼
SOC
▲
|
|
COMMUNICATIONS
```

Why both?

Central Operations needs immediate incident escalation.

Communications provides the technical session/network evidence.

The SOC sits at the intersection.

That's exactly where a security operations function belongs organisationally.

A3.10 But SOC must not simply be an open side of Central Ops

Physically:

```
CENTRAL OPS
|
| glass / security separation
|
├─ Access Reader
|
├─ Secure SOC Door
|
↓
SOC
```

So they are adjacent without being equivalent security zones.

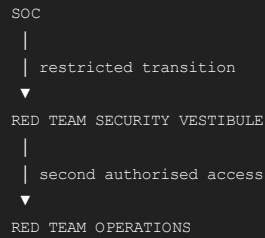
This is an important design pattern we should reuse:

Physical proximity does not mean identical trust.

A3.11 SOC ↔ Red Team

This is another **A — Direct**, but the strongest security transition so far.

I would design:



Not simply:



This makes Incident 05/06 progression much more meaningful.

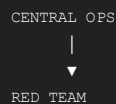
A3.12 Why Red Team should NOT connect directly to Central Operations

This is intentionally:

D — Separate

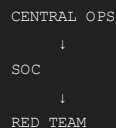
Even though Red Team communicates with HQ Command digitally.

We do not want:



because that destroys the security-depth progression.

Instead:



The spatial architecture says:

Investigation comes before authorised active response.

That matches Incident 06.

A3.13 Red Team should not have casual direct access to Communications either

Also **D — Separate** as a normal player connection.

Although Red Team obviously communicates through Phantom Q infrastructure, it does not need a public doorway directly into Communications.

Operationally:

```
RED TEAM
┆ digital systems
COMMUNICATIONS
```

Spatially:

```
RED TEAM
↓
SOC / controlled operational route
↓
rest of HQ
```

This preserves scope and security.

A3.14 Crypto ↔ SOC

I deliberately set this to:

B — Near

rather than direct.

Because Communications should remain the bridge:

```
CRYPTO
  ↘
  COMMUNICATIONS
    ↘
    SOC
```

That gives the architecture a meaningful information-security chain.

If we add every convenient direct door:

```
Crypto → SOC
Crypto → Central Ops
Crypto → Red Team
```

the facility becomes a web with no hierarchy.

So adjacency must sometimes say **no**, even where a shortcut would be convenient.

A3.15 Engineering ↔ Quantum

This should definitely be:

A — Direct

And importantly, this connection needs to be physically different from ordinary office connections.

Because eventually equipment needs to move between them.

I would plan:

```
ENGINEERING
|
├── Personnel Secure Door
|
├── Equipment-rated controlled opening
|
↓
QUANTUM WING
```

Why?

Engineering may need to support:

- embedded controllers;
- detectors;

- drivers;
- power supplies;
- mounting systems;
- electronics;
- mechanical parts;
- QKD demonstrator components.

We should not be carrying those through Central Operations.

A3.16 Engineering ↔ Central Operations

I put this at:

B — Near

not direct.

Engineering is an important Phantom Q function, but it should feel like the beginning of a technical branch rather than another office attached to the hub.

Conceptually:

```

CENTRAL OPS
|
short technical corridor
|
ENGINEERING
  
```

This creates the transition:

```

OPERATIONS
↓
PHYSICAL ENGINEERING
↓
ADVANCED TECHNOLOGY
  
```

A3.17 Engineering ↔ Foundations

Also **B — Near**.

There is a useful organisational relationship:

```

FOUNDATIONS
education / training
↓
ENGINEERING
practical implementation
  
```

Later that gives us room for practical exercises without putting beginner trainees straight inside Quantum.

They do not need a shared door, but they should not be on opposite ends of the facility.

A3.18 Crypto ↔ Quantum

This is **B — Near**, rather than directly connected.

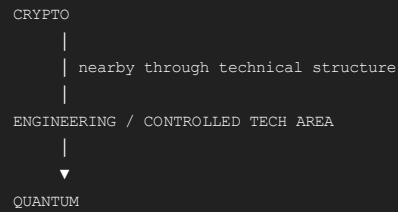
Because there is an intellectual relationship:

```

CLASSICAL CRYPTOGRAPHY
↓
QUANTUM SECURITY
  
```

but Quantum is deliberately deeper in the security/technical hierarchy.

We therefore want:



rather than an ordinary Crypto→Quantum door.

This gives progression physical meaning.

A3.19 Communications ↔ Quantum

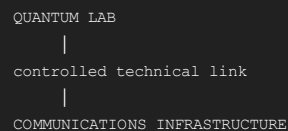
This gets a special **B***.

Because QKD ultimately participates in secure communication.

Technically they should have a strong infrastructure relationship.

But physically, I would avoid a standard public doorway.

Instead there can eventually be:



That may be represented through fibre/cabling/data links rather than player circulation.

This is exactly the difference between:

systems being connected

and

rooms needing to share a door.

A3.20 Advanced Compute requires a different logic

Advanced Compute is not simply another classroom.

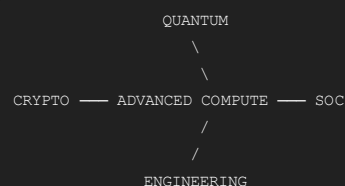
It potentially supports:

SOC analytics
crypto computation
quantum simulation
attack modelling
large simulations
future digital twins

Therefore it should be **near several specialist functions**, but not necessarily directly connected to all of them.

That's why the matrix contains many **B** relationships around Advanced Compute.

Conceptually:



But access remains advanced/restricted.

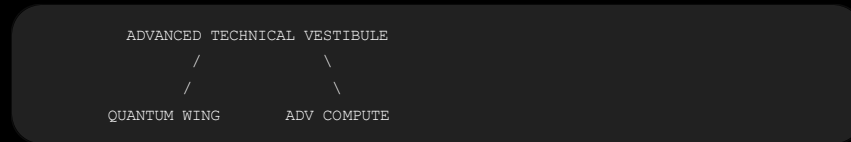
A3.21 Quantum and Advanced Compute

I would currently set:

B — Near

rather than force a direct door.

A strong solution for A4 could be:



So they share a high-security transition zone.

This is cleaner than putting a door directly between the two labs.

A3.22 Secure Core changes the rules

The Secure Core should **not be directly connected to ordinary departments**.

Not Central Ops.

Not SOC.

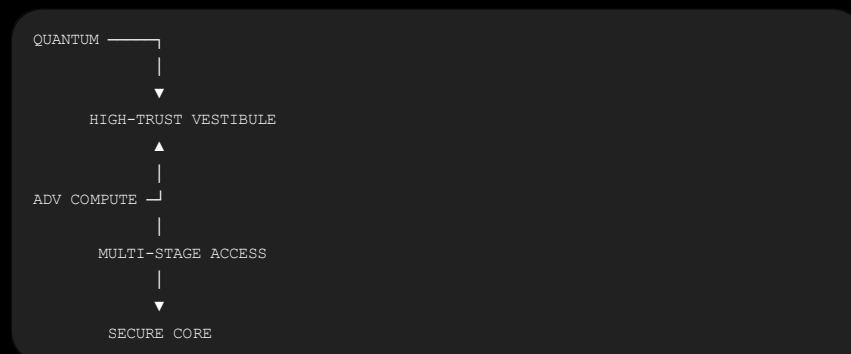
Not Red Team.

Not Crypto.

Not Engineering through normal circulation.

The only sensible approach is a dedicated transition.

Something like:



So in A4 we should probably create a **Secure Core Vestibule**.

It is not another department.

It is a security transition space.

A3.23 Engineering and Secure Core

The matrix marks this **D*** because there is a practical tension.

Operationally:

Engineering may eventually need to maintain equipment that supports the Secure Core.

Security-wise:

Engineering should not have an ordinary direct door into the highest-trust space.

Therefore the future solution should be:

```
ENGINEERING
↓
CONTROLLED SERVICE ROUTE
↓
HIGH-TRUST MAINTENANCE BOUNDARY
↓
SECURE CORE SUPPORT
```

not:

```
Engineering workshop
█
Secure Core
```

This will matter when we design back-of-house circulation.

A3.24 We need transition spaces, not only rooms

A useful result of A3 is that the final building requires several **security transition nodes**.

I would introduce these architectural objects:

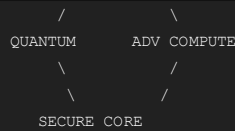
Transition	Between	Function
T0 — HQ Entry Gate	Reception ↔ Central Ops	Identity/access boundary
T1 — SOC Portal	Central Ops/Communications ↔ SOC	Security-operations boundary
T2 — Red Team Vestibule	SOC ↔ Red Team	Authorised active-security boundary
T3 — Technical Gate	Operational core ↔ deeper Engineering zone	Specialist technical boundary
T4 — Quantum/Advanced Vestibule	Engineering ↔ Quantum/Advanced Compute	Advanced technical boundary
T5 — Secure Core Vestibule	Advanced technical zone ↔ Secure Core	Highest-trust boundary

These transitions themselves can become important gameplay landmarks.

A3.25 This creates a progression chain without one linear hallway

We effectively have:

```
FOUNDATIONS
|
ENTRANCE → OPS → COMMUNICATIONS → SOC → RED TEAM
      \      /
      \    /
      CRYPTO
      /
      OPS
      /
      ENGINEERING
      ↓
ADVANCED TECHNICAL GATE
```



So there are really **two major progression branches** beyond Central Ops:

Operational/Security branch

```
OPS
↓
COMMUNICATIONS
↓
SOC
↓
RED TEAM
```

Engineering/Advanced branch

```
OPS
↓
ENGINEERING
↓
QUANTUM / ADVANCED COMPUTE
↓
SECURE CORE
```

And Crypto sits at the intersection of information protection and advanced security.

That gives Phantom Q much more depth than a single level chain.

A3.26 This also solves a future problem

Later campaigns do not have to follow:

```
Incident 07
must unlock the next room after Red Team
```

Because progression can branch.

For example, later a learner could develop:

```
CYBER / SECURITY
```

or:

```
ENGINEERING / QUANTUM
```

or eventually both.

That gives our HQ architecture room for the wider Phantom Q platform.

A3.27 HQ Command / authorisation needs clarification now

Incident 06 repeatedly uses **HQ authority**.

I do **not** think we should create an entirely separate “HQ Department” just to issue signatures.

That would be artificial.

Instead, within or immediately overlooking Central Operations, I recommend eventually defining a restricted:

Command & Authorisation Station

Conceptually:

```
CENTRAL OPERATIONS
|
├─ General operational floor
├─ Workstation 04
├─ Main operations display
└─ COMMAND / AUTHORISATION
    restricted internal sub-zone
```

This gives us a physical home for:

- incident command;
- signed mission orders;
- offensive approvals;
- recovery approvals.

But it remains part of **Central Operations**, not a new department.

That fits our existing architecture better.

A3.28 We also need one non-campaign circulation system

Because the building must be believable, not everything can pass through player-facing corridors.

Our Engineering and Quantum equipment requirements imply a:

technical/service spine

Not necessarily open to the player initially.

Conceptually:

```
SERVICE / EQUIPMENT ENTRY
↓
TECH STORAGE
↓
ENGINEERING
↓
QUANTUM
↓
ADVANCED TECH SUPPORT
```

This means:

- large hardware does not enter through Reception;
- maintenance staff do not roll equipment through Central Ops;
- Quantum equipment has a plausible delivery route.

That is a meaningful architectural requirement for A4.

A3.29 It also helps performance and game design

The service spine can later carry:

- maintenance NPCs;
- equipment deliveries;
- hidden loading;
- utility infrastructure;
- alternate emergency paths.

But we don't need to make it a major Release 1 gameplay corridor.

It simply makes the building make sense.

A3.30 Campaign route validation

Now we check our exact adjacency decisions against the campaign.

Campaign section	Required movement	A3 result
Prologue	Entry → OPS	Direct and logical
Incident 01	OPS → Foundations	Direct
Incident 02	OPS → Crypto/Communications	Short
Incident 03	Crypto ↔ Communications	Direct
Incident 04	OPS → Communications → SOC	Direct escalation chain
Incident 05	SOC ↔ Communications ↔ Crypto; RT standby	Compact cluster + visible deeper RT
Incident 06	SOC ↔ Red Team ↔ HQ authority	Direct RT/SOC + short path back to OPS
Ending	Recovery → Central Ops	Natural return to hub

No unnecessary cross-building routes appear.

That is a strong result.

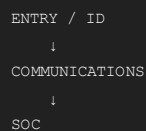
A3.31 Character validation

Alice gets a compact operational triangle:



Excellent.

Eve gets:



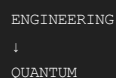
which fits authorised security support without placing her suspiciously inside Red Team.

Red Team gets:



with no inappropriate public shortcut.

Engineering/QKD staff get:



plus a service route.

Again, coherent.

A3.32 Security validation

The player cannot accidentally bypass progression through back doors.

For example, we must avoid:

FOUNDATIONS
[Redacted]
RED TEAM

because that lets physical architecture bypass the entire trust hierarchy.

Similarly:

RECEPTION
[Redacted]
SOC

would make little sense.

And:

CENTRAL OPS
[Redacted]
SECURE CORE

would destroy the entire security-depth concept.

Our D relationships explicitly prevent those mistakes.

A3.33 Handling and camera validation

The adjacency structure also means our commonly used Release 1 rooms cluster around Central Ops:

FOUNDATIONS
CRYPTO
COMMUNICATIONS
SOC
RED TEAM

So the player spends most of Release 1 in a **compact operational area**.

The much larger Engineering/Quantum side can extend deeper without making the early campaign tedious.

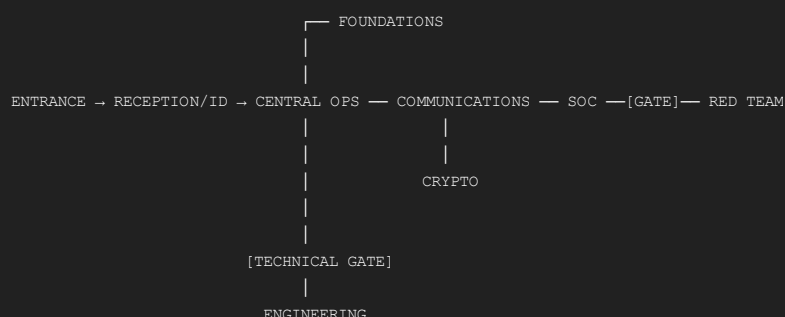
That's good for both:

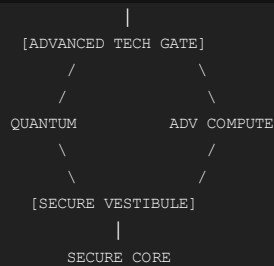
- handling;
- rendering.

Early builds can have the advanced wing present in greybox/locked form without requiring full visual detail.

A3.34 The adjacency graph I would now carry into A4

This is the most useful condensed version:





I would **not yet call that the final floor plan**.

But this is now strong enough that A4 has constraints.

A3 — What is now effectively locked

The major direct adjacencies are:

Entry ↔ Central Operations, Central Ops ↔ Foundations, Central Ops ↔ Communications, Central Ops ↔ SOC, Communications ↔ Crypto, Communications ↔ SOC, SOC ↔ Red Team, and Engineering ↔ Quantum.

The most important deliberate separations are equally important: **Reception must not directly access sensitive departments; Central Ops must not directly access Red Team, Quantum, Advanced Compute or Secure Core; Red Team must sit behind SOC rather than beside normal operations; Secure Core must have no ordinary departmental doorway.**

We have also discovered two architectural requirements that were not obvious when we started: the HQ needs **security transition/vestibule spaces**, and the Engineering/Quantum side needs a **technical service/equipment route** so that the building works as an engineering facility rather than just a game level.

That means A1 → A2 → A3 still loops cleanly. We have not yet found a reason to break the central-hub concept.

A4 is now the important one: Whole HQ Floor Plan. At that point we can finally convert these relationships into actual geometry — building footprint, wings, room positions, corridor geometry, security vestibules, service route and approximate dimensions — and then loop the resulting floor plan back through A1–A3 before we accept it.